

Customer Shopping Behavior Analysis

Project Report | Python · SQL · Power BI

Objective: use 3,900 customer purchases to find spending patterns, customer segments and loyalty drivers, and turn them into marketing and product recommendations.

Executive Summary

- **\$233,081** total revenue from 3,900 customers; the average purchase is \$59.76 and the average review rating is 3.75.
- **Clothing leads** with 44.7% of revenue; Accessories add 31.8%. Together they account for over three quarters of sales.
- **Discounts and subscriptions do not raise spend:** discounted purchases average \$59.28 vs \$60.13 without, and subscribers spend \$59.49 vs \$59.87.
- **The base is loyal but not subscribed:** 80% of customers have 11+ previous purchases, yet only 27% subscribe, and repeat buyers subscribe at the same rate (27.6%).
- **Men generate 68% of revenue** because they are 68% of customers; average spend per purchase is almost the same for women (\$60.25) and men (\$59.54).

1. Project Overview

This project analyzes transactional data from 3,900 purchases across four product categories. It follows the full analytics workflow: data preparation in Python, business analysis in SQL and an interactive Power BI dashboard, ending in recommendations for management.

2. Dataset Summary

Item	Raw data	Cleaned data
Rows	3,900	3,900
Columns	18	19
Missing values	37 (Review Rating)	0
Duplicate customer IDs	None	None
Customer age range	18–70	18–70

Key features cover customer demographics (age, gender, location, subscription status), purchase details (item, category, amount, season, size, color) and shopping behavior (discount applied, previous purchases, purchase frequency, review rating, shipping type, payment method).

3. Data Preparation in Python

- **Loading and exploration:** imported the data with pandas and reviewed structure with `df.info()` and `df.describe()`.
- **Missing data:** 37 missing Review Rating values were filled with the median rating of each product category.
- **Column standardization:** all column names converted to snake_case.
- **Age groups:** age_group created from four bands: Young Adult (18–31), Adult (32–44), Middle Age (45–57) and Senior (58–70).
- **Purchase frequency:** purchase_frequency_days created by converting the frequency labels into days (Weekly = 7, Fortnightly/Bi-Weekly = 14, Monthly = 30, Quarterly/Every 3 Months = 90, Annually = 365).
- **Redundancy check:** discount_applied and promo_code_used carried the same information, so promo_code_used was dropped.
- **Database load:** the cleaned data (customer_shopping_cleaned.csv) was loaded into a SQL database for analysis.

4. Data Analysis in SQL

Sixteen queries answer the key business questions. Results are summarized below; the full queries are in customer_shopping_sql_queries.sql.

Q1. Revenue by gender

What is the total revenue from male vs. female customers?

Gender	Customers	Revenue (\$)	Share
Male	2,652	157,890	67.7%
Female	1,248	75,191	32.3%

Takeaway: Revenue split mirrors the customer mix (68% male). Average spend per purchase is nearly equal: \$59.54 male vs \$60.25 female.

Q2. High-spending discount users

Which customers used a discount but still spent more than the average purchase (\$59.76)?

Measure	Value
Purchases with a discount	1,677
Of these, above the average purchase	839
Share	50.0%

Takeaway: Half of discounted purchases were still above average, so discounts are not limited to small baskets.

Q3. Top 5 products by rating

Which products have the highest average review rating?

Product	Avg. rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Takeaway: Ratings are tightly bunched (3.78–3.86), so review score alone will not separate winners; use it alongside sales.

Q4. Shipping type comparison

How do average purchase amounts compare between Standard and Express shipping?

Shipping type	Avg. purchase (\$)
Express	60.48
Standard	58.46

Takeaway: Express buyers spend about \$2.02 (3.4%) more per purchase, a small but consistent gap.

Q5. Subscribers vs. non-subscribers

Do subscribed customers spend more?

Subscribed	Customers	Avg. spend (\$)	Total revenue (\$)
Yes	1,053	59.49	62,645
No	2,847	59.87	170,436

Takeaway: Subscribers do not spend more per purchase, and they are only 27% of customers and 27% of revenue.

Q6. Discount-dependent products

Which 5 products have the highest share of discounted purchases?

Product	Discount rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Takeaway: Discount use is about 43% overall (1,677 of 3,900); these products sit at 47–50%, so the spread is modest.

Q7. Customer segmentation

Segment customers by previous purchases: New (1), Returning (2–10), Loyal (11+).

Segment	Customers	Share
Loyal	3,116	79.9%
Returning	701	18.0%
New	83	2.1%

Takeaway: The base is highly loyal, but very few new customers are entering, which is a growth risk.

Q8. Top products per category

What are the most purchased products in each category?

Rank	Category	Product	Orders
1	Accessories	Jewelry	171
2	Accessories	Belt	161
3	Accessories	Sunglasses	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Takeaway: Orders are evenly spread (145–171 per product). Outerwear has only two products, so it lists two rows.

Q9. Repeat buyers and subscriptions

Are customers with more than 5 previous purchases more likely to subscribe?

Subscribed	Repeat buyers	Share
No	2,518	72.4%
Yes	958	27.6%

Takeaway: No. Repeat buyers subscribe at 27.6%, about the same as all customers (27.0%). Loyalty is not turning into subscriptions.

Q10. Revenue by age group

What is the revenue contribution of each age group?

Age group	Revenue (\$)	Share
Young Adult (18–31)	62,143	26.7%
Middle Age (45–57)	59,197	25.4%
Adult (32–44)	55,978	24.0%
Senior (58–70)	55,763	23.9%

Takeaway: Revenue is spread fairly evenly; Young Adults lead by about \$6K over the smallest group.

Q11. Revenue by category

Which categories generate the most revenue?

Category	Revenue (\$)	Share
Clothing	104,264	44.7%
Accessories	74,200	31.8%
Footwear	36,093	15.5%
Outerwear	18,524	7.9%

Takeaway: Clothing and Accessories bring 76.5% of revenue.

Q12. Revenue by season

Which season has the highest revenue and average purchase?

Season	Revenue (\$)	Avg. purchase (\$)
Fall	60,018	61.56
Spring	58,679	58.74
Winter	58,607	60.36
Summer	55,777	58.41

Takeaway: Fall is the strongest season on both measures; Summer is the weakest.

Q13. Payment methods

What share of customers use each payment method?

Method	Customers	Share
PayPal	677	17.4%
Credit Card	671	17.2%
Cash	670	17.2%
Debit Card	636	16.3%
Venmo	634	16.3%
Bank Transfer	612	15.7%

Takeaway: No method dominates; customers are split almost evenly, so all six should stay available.

Q14. Top locations

Which locations have the most customers and revenue? (top 5 shown)

Location	Customers	Revenue (\$)
Montana	96	5,784
California	95	5,605
Idaho	93	5,587
Illinois	92	5,617
Alabama	89	5,261

Takeaway: Demand is spread across 50 locations; the top five differ by only a few customers.

Q15. Effect of discounts

Does applying a discount change the average purchase amount?

Discount applied	Purchases	Avg. purchase (\$)
No	2,223	60.13
Yes	1,677	59.28

Takeaway: Discounted purchases are slightly smaller (\$0.85 lower), so discounts are not lifting basket size.

Q16. Product revenue rank within category

Rank products by total revenue within each category (top 2 shown).

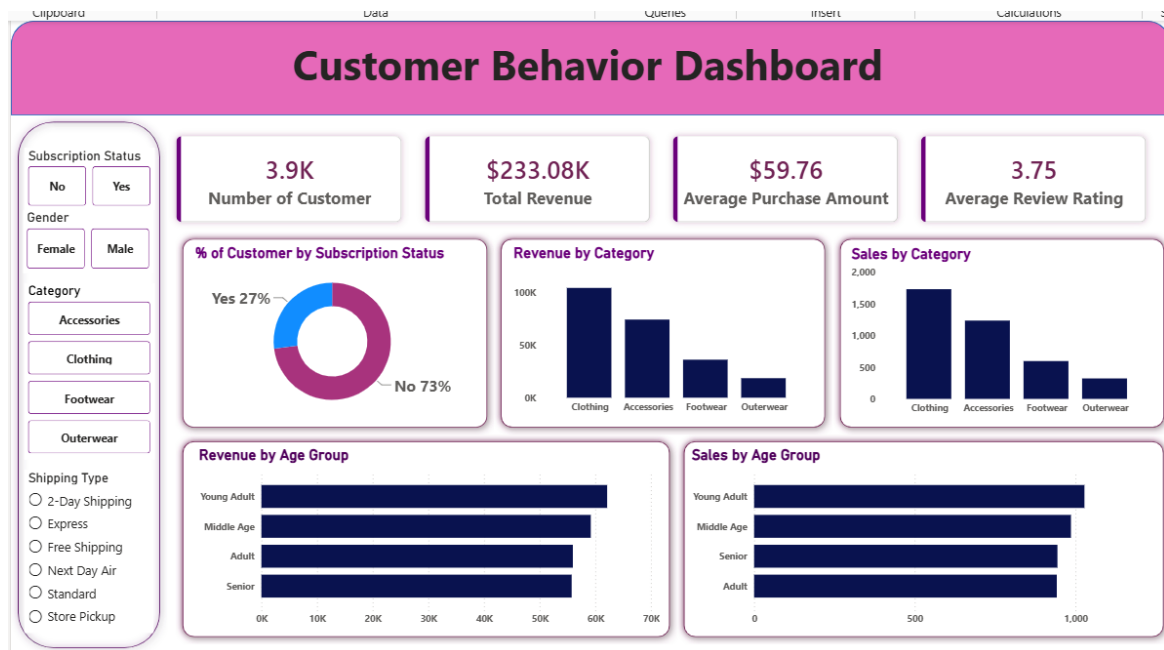
Category	Rank	Product	Revenue (\$)
Accessories	1	Jewelry	10,010
Accessories	2	Sunglasses	9,649
Clothing	1	Blouse	10,410
Clothing	2	Shirt	10,332
Footwear	1	Shoes	9,240
Footwear	2	Sandals	9,200
Outerwear	1	Coat	9,275
Outerwear	2	Jacket	9,249

Takeaway: Blouse (\$10,410) is the top single product by revenue, followed by Jewelry (\$10,010).

5. Dashboard in Power BI

The interactive dashboard shows headline KPIs (3.9K customers, \$233.08K revenue, \$59.76 average purchase, 3.75 average rating) plus subscription mix, revenue and sales by

category, and revenue and sales by age group. Filters let stakeholders slice by subscription status, gender, category and shipping type.



6. Key Insights

- Category concentration: Clothing and Accessories drive 76.5% of revenue; Outerwear is only 7.9%.
- Discounts are not paying for themselves: discounted purchases are slightly smaller, yet 43% of purchases use one.
- Subscriptions are under-used: subscribers spend the same as non-subscribers and make up 27% of customers, even among repeat buyers.
- Loyalty is high, acquisition is low: 80% of customers are Loyal and only 2% are New.
- Gender gap is a reach gap: women spend as much per purchase as men but are only 32% of customers.
- Seasonality is mild: Fall is the peak (\$61.56 average) and Summer the low (\$58.41).

7. Business Recommendations

Recommendation	Evidence	Action
Make subscriptions worth having	27% subscribe; repeat buyers 27.6%; no spend lift	Add real perks (early access, free shipping) and offer them to Loyal customers
Reward loyalty to lift spend	80% Loyal, yet spend is flat	Tiered rewards that unlock at higher basket values
Tighten discount policy	Discount purchases avg. \$59.28 vs \$60.13	Target discounts at New and lapsed customers; protect margin elsewhere
Acquire new customers	Only 83 New customers (2.1%)	Run acquisition campaigns and first-purchase offers
Reach more women	32% of customers, equal spend	Test targeted campaigns in Clothing and Accessories
Lead campaigns with top products	Clothing 44.7%; Blouse, Jewelry top revenue	Feature best sellers and top-rated items (Gloves, Sandals, Boots)
Plan around Fall	Highest revenue and average purchase	Build inventory and promotions ahead of Fall; run Summer offers
Promote Express shipping	Express buyers spend \$2.02 more	Highlight Express at checkout for larger baskets

8. Limitations

- The data has no online vs. offline channel field, so channel trends could not be analyzed.
- Each row is one purchase per customer, so customer-level repeat behavior relies on the `previous_purchases` field.
- Differences in spend across discounts, subscriptions, shipping and age groups are small; treat them as directional and confirm with further testing.